Shenzhen Damiao Technology Co., Ltd.

Motor Module Selection Guide

Type	Motor Model	Rated Torque (Nm)	Peak Torque (Nm)	No-load Speed (rpm)	Rated Speed (rpm)	Gear Ratio	Dimensions Diameter*Height (mm)	Weight (g)	Supply Voltage (V)	Recommended operating Voltage Range (V)	Rated Phase Current (A)	Peak Phase Current (A)	Rated Power (W)	Pole Pairs	Slots	Communication Interface	Encoder Type	Drive Configuration	Phase-to-Phase Resistance (Ω)	Phase-to-Phase Inductance (uH)	Flux Linkage (Wb)	Rotor Inertia (Kg*m^2)	Torque Ratio	Torque Constant (Nm/A)	Maximum Drive Current (A)	Default PMAX (rad)	Default VMAX (rad/s)	Default TMAX (Nm)	Speed Constant(Kv) (rpm/V)
Joint Motor DM - [J + Stator Diameter + Gear Ratio] - [Encoder(1E/2E) + Communication Interface]	DM-J3507-2EC	0.8	3	460	150	7	46*37.9	≈150	24	15-30	3	8.3	12.6	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.8	235	0.0037	8.70E-06	1	0.5439	10.261194	12.566	50	5	106.4341497
	DM-J3507-2EC （48V）	0.8	3	910	150	7	46*37.9	≈150	48	15-52	3	8.3	12.6	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.8	235	0.0037	8.70E-06	1	0.5439	10.261194	12.566	50	5	106.4341497
	DM-J4310-2EC V1.1	3	7	200	120	10	57*46	≈306	24	15-32	2.5	7.5	37.7	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.65	340	0.0045	1.80E-05	1	0.945	10.261194	12.5	30	10	87.51252308
	DM-J4310-2EC V1.1 （48V）	3	7	400	120	10	57*46	≈306	48	15-52	2.5	7.5	37.7	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.65	340	0.0045	1.80E-05	1	0.945	10.261194	12.5	30	10	87.51252308
	DM-J4310-2EC V1.2	3.5	12.5	200	120	10	57*46	≈306	24	15-32	4.9	20	44.0	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.58	320	0.0044	2.00E-05	1	0.924	20.522388	12.5	30	10	89.50144406
	DM-J4310-2EC V1.2 （48V）	3.5	12.5	450	120	10	57*46	≈306	48	15-52	4.8	20	44.0	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.58	320	0.0044	2.00E-05	1	0.924	20.522388	12.5	30	10	89.50144406
	DM-J4310P-2EC	3.5	12.5	200	120	10	57*49.05	≈330	24	15-32	4.9	20	44.0	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.58	320	0.0042	1.90E-05	1	0.882	20.522388	12.5	30	10	93.76341759
	DM-J4310P-2EC （48V）	3.5	12.5	450	120	10	57*49.05	≈330	48	15-52	4.8	20	44.0	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.58	320	0.0042	1.90E-05	1	0.882	20.522388	12.5	30	10	93.76341759
	DM-J4340-2EC	9	27	52.5	36	40	57*53.3	≈362	24	15-32	3	8	33.9	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.88	360	0.00485	2.00E-05	1	4.074	10.261194	12.5	8	28	81.19718637
	DM-J4340-2EC （48V）	9	27	100	36	40	57*53.3	≈362	48	15-52	2.5	9	33.9	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.88	360	0.00485	2.00E-05	1	4.074	10.261194	12.5	8	28	81.19718637
	DM-J4340-2EC V1.1	12	40	56	36	40	57*53.3	≈362	24	15-32	4.11	19.85	45.2	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.72	290	0.0043	1.97E-05	1	3.612	20.522388	12.5	10	28	91.58287299
	DM-J4340-2EC V1.1 （48V）	12	40	112	36	40	57*53.3	≈362	48	15-52	4.11	19.85	45.2	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.72	290	0.0043	1.97E-05	1	3.612	20.522388	12.5	10	28	91.58287299
	DM-J4340P-2EC	9	27	52.5	36	40	57*56.5	≈381	24	15-32	3	8	33.9	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.88	360	0.00485	2.00E-05	1	4.074	10.261194	12.5	8	28	81.19718637
	DM-J4340P-2EC （48V）	9	27	100	36	40	57*56.5	≈381	48	15-52	2.5	9	33.9	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.88	360	0.00485	2.00E-05	1	4.074	10.261194	12.5	8	28	81.19718637
	DM-J4340P-2EC V1.1	12	40	56	36	40	57*56.5	≈381	24	15-32	4.11	19.85	45.2	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.72	290	0.0043	1.97E-05	1	3.612	20.522388	12.5	10	28	91.58287299
	DM-J4340P-2EC V1.1 （48V）	12	40	112	36	40	57*56.5	≈381	48	15-52	4.11	19.85	45.2	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.72	290	0.0043	1.97E-05	1	3.612	20.522388	12.5	10	28	91.58287299
	DM-J6006-2EC	4	11	226@24V 408@48V	150	6	76*36.5	≈335	24-48	15-52	5.96	17.6	62.8	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.435	270	0.0058	5.80E-05	1	0.7308	20.52238806	12.5	45	20	67.89764722
	DM-J6248P-2EC	30	97	60@24V 115@48V	40	48	76*62.5	≈660	24-48	15-65	11.3	38.7	125.7	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.17	85	0.0033	5.40E-05	1	3.3264	41.044777	12.5	20	120	119.3352587
	DM-J8006-2EC V1.1	8	20	194.2@24V 392.8@48V	120	6	96*40	≈559	24-48	15-52	9	21	100.5	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.4	215	0.0054	1.15E-04	1	1.0206	41.04477612	12.5	45	40	48.61806838
	DM-J8009-2EC	20	40	168@24V	100@24V	9	98*61.7	≈896	24-48	15-52	18.6	40	209.4	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
		20	40	335@48V	200@48V	9	98*61.7	≈896	24-48	15-52	18.6	40	418.9	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
	DM-J8009P-2EC	20	40	168@24V	100@24V	9	98*61.7	≈963	24-48	15-52	18.6	40	209.4	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
		20	40	335@48V	200@48V	9	98*61.7	≈963	24-48	15-52	18.6	40	418.9	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
	DM-J8009P-2EC V1.1	20	40	168@24V	100@24V	9	98*61.7	≈950	24-48	15-52	18.6	40	209.4	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
		20	40	335@48V	200@48V	9	98*61.7	≈950	24-48	15-52	18.6	40	418.9	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.145	80	0.00445	1.95E-04	1	1.261575	41.04477612	12.5	45	54	58.99720657
	DM-J10010-2EC	40	150	75@24V	50@24V	10	112*62 (Max. Overall Dimension: 120)	≈1485	24-48	15-52	18.4	80.2	209.4	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.125	100	0.00825	5.50E-04	1	2.59875	100	12.5	20	200	31.82273567
		40	150	150@48V	100@48V	10	112*62 (Max. Overall Dimension: 120)	≈1485	24-48	15-52	18.4	80.2	418.9	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.125	100	0.00825	5.50E-04	1	2.59875	100	12.5	20	200	31.82273567
	DM-J10010L-2EC	40	120	100@24V	70@24V	10	120*53 Height (Excluding Dowel Pin)	≈1375	24-48	15-52	23.5	95	293.2	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.11	85	0.00635	5.56E-04	1	2.00025	99.7	12.5	25	200	41.34449909
		40	120	200@48V	100@48V	10	120*53 Height (Excludingt Locating Pin)	≈1375	24-48	15-52	23.5	95	418.9	21	36	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.11	85	0.00635	5.56E-04	1	2.00025	99.7	12.5	25	200	41.34449909
	DM-J10422P-2EC	100	400	120	100	22	110*89.9	≈2700	48	20-65	44.5	165	1047.2	16	36	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.081	38	0.00599	5.64E-04	1	3.16272	165	12.566	20	500	57.52596989
	DM-J8520P-2EC	50	176	124	120	20	90*70	≈1392	48	20-65	13.86	41.6	628.3	16	32	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.1	75	0.00651	1.49E-04	1	3.38976	82	12.566	20	300	53.67298437
	DM-J6216L-2EC	10	45	260	200	16	67*62.5	≈740	24-48	15-52	7.48@24V 7.74@48V	39.5@24V 27@48V	209.4	10	24	CAN、CANFD	Inductive +Magnetic Encoder Dual Encoder	Integrated	0.16	184	0.00510	4.91E-05	1	1.224	82.5	12.566	40	70	108.103705
	DM-J6216P-2EC	10	45	260	200	16	67*62.5	≈740	24-48	15-52	20.8@24V 22@48V	36@24V 70@48V	209.4	10	24	CAN、CANFD	Inductive +Magnetic Encoder Dual Encoder	Integrated	0.16	184	0.00510	4.91E-05	1	1.224	82.5	12.566	40	70	108.103705
	DM-J4310M-2EC	5.2	18	232	120	10	57*53.3	≈ 370	24	15-32	8.49	32	65.3	14	24	CAN、CANFD	Magnetic Encoder Dual Encoder	Integrated	0.327465	210.6415	0.00418	2.16E-05	1	0.8778	41.044777	12.5	30	20	94.21204638
Separable Motor DM - [S + Stator Diameter + Gear Ratio] - [Encoder(1E/2E) + Communication Interface]	DM-S3519-1EC	3.5	7.8	435	395	19.2032	42*91.5	≈396	24	15-32	9.2	20.5	144.8	7	12	CAN、CANFD	Hall Encoder Incremental Encoder	separate	0.2	55	0.00225	1.80E-05	1	0.4536756	20.52238806	12.5	200	10	350.0500923
	DM-S2325-1EC	1.35	5	560	380	25	28*77.6	≈172	24	15-60	5	13	53.7	7	12	CAN、CANFD	Hall Encoder Incremental Encoder	separate	0.292	75	0.0014	3.80E-05	1	0.3675	20.523388	12.5	200	10	562.5805055

Type	Motor Model	Rated Torque (Nm)	Peak Torque (Nm)	No-load Speed (rpm)	Rated Speed (rpm)	Gear Ratio	Dimensions Diameter*Height (mm)	Weight (g)	Supply Voltage (V)	Recommended operating Voltage Range (V)	Rated Phase Current (A)	Peak Phase Current (A)	Rated Power (W)	Pole Pairs	Slots	Communication Interface	Encoder Type	Drive Configuration	Phase-to-Phase Resistance (Ω)	Phase-to-Phase Inductance (uH)	Flux Linkage (Wb)	Rotor Inertia (Kg*m^2)	Torque Ratio	Torque Constant (Nm/A)	Maximum Drive Current (A)	Default PMAX (rad)	Default VMAX (rad/s)	Default TMAX (Nm)	Speed Constant(Kv) (rpm/V)
Hub Motor DM - [H + Stator Diameter + Height]	DM-H6215	1	2	320	120	1	68*42.5	≈360	24	15~32	2.5	4.3	12.6	14	24	CAN、CANFD	Hall Encoder Incremental Encoder	Integrated	2.5	2000	0.022	1.80E-04	1	0.462	10.261194	12.5	30	10	17.90028881
	DM-H3510	0.1	0.454	1700	500	1	42*31.5	≈116	24	15~32	0.79	1.88	5.2	14	24	CAN、CANFD	Magnetic Encoder Absolute	Integrated	2.76	470	0.011	1.74E-05	1	0.231	10.261194	12.5	280	1	35.80057762
	DM-H65	6	21.5	260@24V 395@36V 480@48V	120	1	171*66.3	≈3150	24-48	20~65	9.4	33.6	250 [6 Nm, 400 rpm @ 48V (S2 Duty)]	15	27	CAN、CANFD	Inductive Encoder Absolute	external	0.42	750	0.024964	1.68E-03	1	0.56169	41.044777	12.566	100	40	14.72330544
	DM-H55	3.5	9	587@24V 883@36V 1185@48V	350	1	141*66.9	≈2100	24-48	20-65	13.05	33.55	128.3	10	18	CAN、CANFD	Inductive Encoder Absolute	external	0.281	336	0.018163	2.07E-03	1	0.272445	41.04477	12.566	100	40	30.35450616
Hollow Gimbal DM - [G + Stator Diameter + Height]	DM-G6220	1.3	2.7	300	110	1	68*41.5	≈493.5	24	15~32	2.3	4.6	15.0	14	24	CAN、CANFD	Magnetic Encoder Absolute	Integrated	3	3000	0.028	1.25E-04	1	0.588	10.261194	12.5	45	10	14.06451264
	DM-G6220（48V）	1.3	4.9	650	110	1	68*41.5	≈493.5	48	15~52	0.8	7.82	15.0	14	24	CAN、CANFD	Magnetic Encoder Absolute	Integrated	2.84	2376	0.028	1.25E-04	1	0.588	10.261194	12.5	45	10	14.06451264
	DM-G8518	1.9	8.78	1290@24V	1000@24V	1	96*52.5	≈831	24-48	20-54	9.55A	36.1	199.0	16	36	CAN、CANFD	Inductive Encoder Absolute	Integrated	0.225	105	0.0065	1.86E-04	1	0.156	82.089554	12.566	300	15	53.01239379
		1.9	8.78	2590@48V	2000@48V	1	96*52.5	≈831	24-48	20-54	9.55A	36.1	397.9	16	36	CAN、CANFD	Inductive Encoder Absolute	Integrated	0.225	105	0.0065	1.86E-04	1	0.156	82.089554	12.566	300	15	53.01239379
Harmonic Drive joint motor DM - [JH + Size] - [Gear Ratio] - [Encoder(1E/2E) + Communication Interface]	DM-JH11-51-2EC	3.2	7.8	60	40	51	52*60	≈345	24-48	20~54	2.23	4.26	13.4	7	12	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	1.1	538	0.0048	6.30E-06	0.6	1.54224	10.261194	12.5	10	12	164.0859808
	DM-JH11-101-2EC	4.8	10.5	30	20	101	52*60	≈345	24-48	20~54	2.3	3.6	10.1	7	12	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	1.1	538	0.0048	6.30E-06	0.65	3.30876	10.261194	12.5	10	12	164.0859808
	DM-JH14-51-2EC	5.4	18	60	40	51	70*70	≈775	24-48	20~54	4.89	13.7	22.6	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.243	170	0.0053518	6.00E-05	0.5	2.04707498	41.044777	12.566	10	50	103.0168924
	DM-JH14-101-2EC	7.8	28	30	20	101	70*70	≈775	24-48	20~54	4.12	10.97	16.3	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.243	170	0.0053518	6.00E-05	0.45	3.6486101	41.044777	12.566	10	50	103.0168924
	DM-JH17-51-2EC	16	24	60	40	51	80*70	≈985	24-48	20~54	4.62	12.94	67.0	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.161	107	0.00625	9.38E-05	0.5	4.78125	41.044777	12.566	20	80	88.21262327
	DM-JH17-101-2EC	24	54	30	20	101	80*70	≈985	24-48	20~54	6.31	16.85	50.3	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.161	107	0.00625	9.38E-05	0.5	9.46875	41.044777	12.566	20	80	88.21262327
	DM-JH20-51-2EC	32	69	60	40	51	90*73	≈1428	48	20~54	6.42@24V 6.3@48V	13.2@24V 13.7@48V	134.0	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.2	163.3	0.00325	5.47E-05	1	2.48625	41.044777	12.566	10	85	169.6396601
	DM-JH20-101-2EC	50	102	30	20	101	90*73	≈1434	48	20~54	3.52@24V 3.48@48V	10.3@24V 10.05@48V	104.7	10	18	CAN、CANFD	Inductive Encoder Dual Encoder	Integrated	0.2	163.3	0.00325	5.47E-05	1	4.92375	41.044777	12.566	10	85	169.6396601
Direct Drive Motor DM - [D + Stator Diameter + Height] - [Encoder(1E/2E)+ Communication Interface]	DM-D5730 -1EE	0.65	1.9	1500	1100	1	57*95.2	≈628	24-48	20~55	5.4	16.2	74.9	14	24	CAN、CANFD EtherCAT	Inductive Encoder Absolute	Integrated	0.17	85	0.00617	6.00E-05	1	0.12957	20.522388	12.5	420	5	63.82598928
	DM-D5730 -1EC	0.65	1.9	1500	1100	1	57*95.2	≈628	24-48	20~55	5.4	16.2	74.9	14	24	CAN、CANFD	Inductive Encoder Absolute	Integrated	0.17	85	0.00617	6.00E-05	1	0.12957	20.522388	12.5	420	5	63.82598928

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